

Karthik.K

Software Engineer (Web & AI/ML)

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Profile Summary

A skilled **Software Engineer (Web & AI/ML)** with a strong foundation in front-end and back-end development, specializing in creating dynamic, responsive, and user-friendly web applications. Proficient in **HTML, CSS, JavaScript, React**, and **Node.js**, with expertise in building scalable and maintainable solutions. Demonstrates a deep understanding of full-stack development—including database integration and API creation—to deliver seamless user experiences. Possesses foundational **AI/ML** knowledge in data preprocessing, model evaluation, and recommendation systems using Python libraries such as scikit-learn and pandas. Passionate about coding, problem-solving, and staying updated with emerging technologies to deliver innovative solutions. Known for effective communication, teamwork, and a results-driven approach to achieving project goals.

Technical Skills

Programming Languages: Python, Core Java, C.

AI & ML: Data preprocessing, Model evaluation, Basic use of scikit-learn, pandas.

Front-End Development : HTML5, CSS3, JavaScript, Bootstrap, Tailwind CSS.

Back-End Development: Node.js.

Database Management: MySQL, MongoDB, Firebase.

Frameworks and Libraries: React.js, Angular, Express.js.

Version Control: Git, GitHub.

Projects

Chat Application (MERN Stack)

A feature-rich real-time chat application designed for seamless communication.

- **Frontend:** Developed using React with ShadCN for modular UI components and styled with Tailwind CSS for a responsive, visually appealing design.
- **Authentication:** Secured with JWT tokens for user verification.
- **File Handling:** Integrated Multer for file/image uploads and downloads.
- **Real-Time Messaging:** Enabled instant communication using sockets, supporting both group chats and personal messages with emoji integration.
- **Backend:** Built with Node.js and Express for scalable and efficient server-side functionality.
- **Database:** Utilized MongoDB for reliable and efficient data storage.
- **State Management:** Implemented Zustand for effective client-side state management.
- **API Interaction:** Facilitated smooth API calls using Axios.
- **Code Quality:** Followed best practices to ensure structured, maintainable code.

Portfolio Website

[Portfolio](#)

A personal portfolio website showcasing projects, skills, and achievements, designed to highlight expertise and creativity.

- **Frontend:** Built using HTML, CSS, and JavaScript for an interactive and visually appealing design.
- **Styling:** Tailored with CSS for sleek and modern aesthetics.
- **Responsiveness:** Fully responsive design ensuring compatibility across devices.
- **Navigation:** Clean and intuitive layout for seamless user experience.
- **Deployment:** Hosted on GitHub Pages for easy access and high reliability.
- **Code Quality:** Developed with clean, structured, and maintainable code.

Mini Projects

Real-Time Weather Application

[Live-Link](#)

A dynamic application providing real-time weather updates with location-based accuracy.

- **Frontend:** Developed using HTML, CSS, and JavaScript for an interactive and user-friendly interface.
- **Geolocation API:** Utilized to automatically detect user location for accurate weather information.
- **API Integration:** Integrated with OpenWeatherMap API to fetch real-time weather data such as temperature, conditions, and forecasts.
- **Styling:** Designed with responsive CSS for compatibility across devices.
- **Features:** Supports manual location input and displays weather details based on current or selected location.
- **Deployment:** Hosted on GitHub Pages for accessible online usage.

Basic Calculator

[Live-Link](#)

A functional calculator application for performing basic arithmetic operations.

- **Frontend:** Built using HTML and CSS for a clean and user-friendly interface.
- **Functionality:** Implemented JavaScript for core arithmetic operations, including addition, subtraction, multiplication, and division.
- **Responsiveness:** Designed with CSS to ensure compatibility with various devices.
- **User Experience:** Includes a clear display and intuitive button layout for ease of use.
- **Deployment:** Hosted on GitHub Pages for seamless accessibility.

- **Frontend:** Built using HTML and CSS for a visually engaging and responsive interface.
- **Functionality:** Implemented game logic using JavaScript to simulate dice rolls and determine winners based on random outcomes.
- **User Interaction:** Features intuitive controls and dynamic updates to enhance player experience.
- **Styling:** Designed with CSS for a clean and attractive layout.
- **Deployment:** Hosted on GitHub Pages for easy online access.

Student Education Recommendation System (using AI)
A personalized course and pathway recommender using AI.

- **Tech & Tools:** Python, pandas, scikit-learn, Flask
- **Data Pipeline:** Cleaned and transformed student survey data; engineered features like GPA trends & interest scores
- **AI Approach:** Collaborative filtering (k-NN) and content-based filtering (cosine similarity)
- **Demo:** Deployed as a Flask app where students input their profile and receive tailored course suggestions

Certifications and Achievements

- Deloitte Australia Technology Job Simulation on Forage - June 2025**
- [Certificate](#)
- Completed a job simulation involving development and coding
 - Wrote a proposal for creating a dashboard

Education

Bachelor of Technology in Information Technology	2022-2026
Er.Perumal Manimekalai College Of Engineering.	
Higher Secondary Education (+2)	2020-2022
SRI VIJAY VIDHYALAYA HIGHER SECONDARY SCHOOL, HOSUR.	
CGPA: 9.5/10.0	